SOFTWARE DESIGN REVIEW

SUMMARY OF TOPICS

- Introduction
- Class&Package Design Principles
- Architectural Characteristics and Component-based thinking
- Architectural Styles
- Design Patterns

INTRODUCTION

OOP review

- Classes
- Objects
- Relationships: inheritance, composition
- Abstract classes, Interfaces
- Polymorphism

CLASS & PACKAGE PRINCIPLES

- SOLID
- GRASP
- Package-related Cohesion Principles
- Package-related Coupling Principles

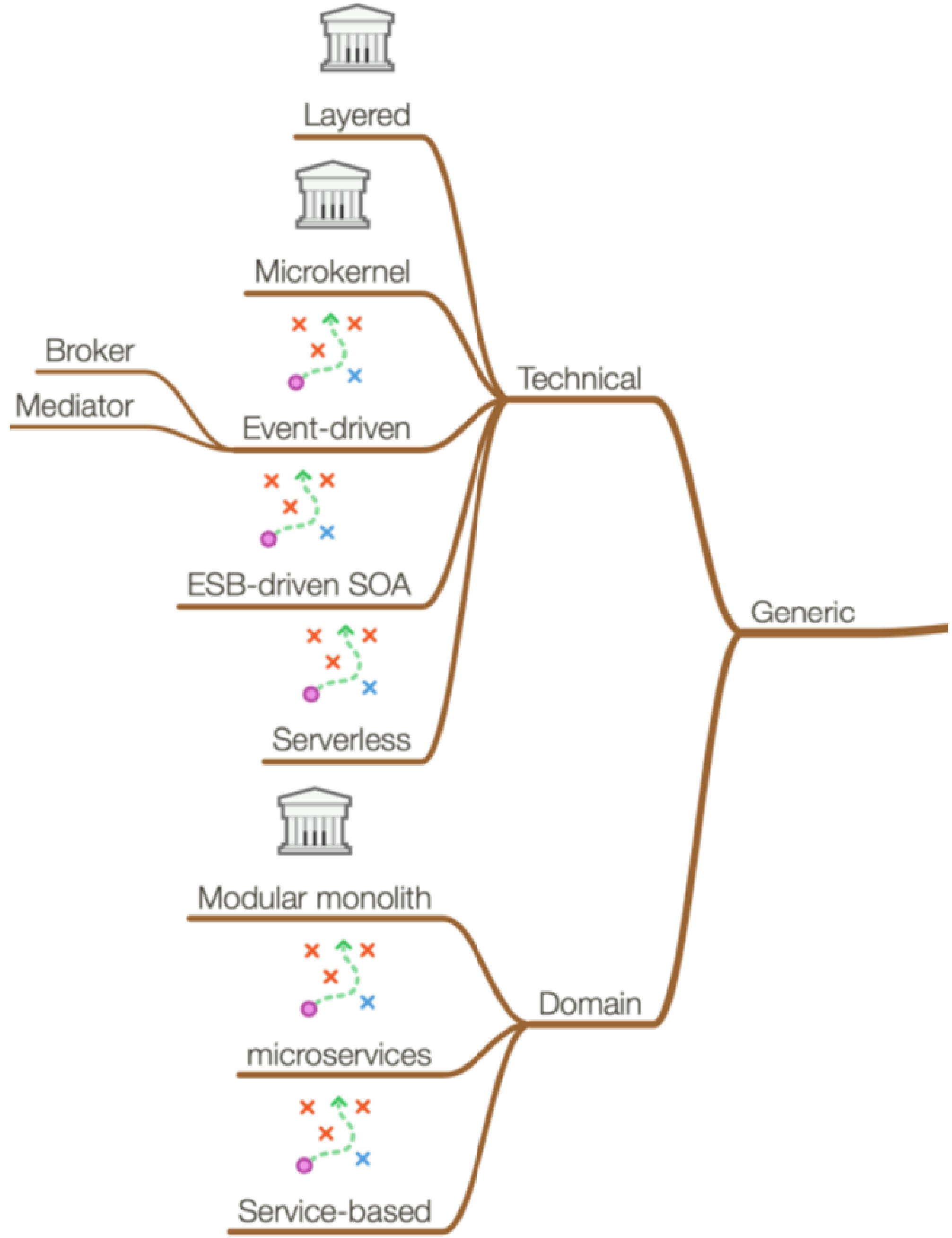
References

- Robert Martin Articles
- Solid eBook http://cdn.cloudfiles.mosso.com/c82752/pablos_solid_ebook.pdf
- Craig Larman, *Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development*, 3rd Ed, Addison Wesley, 2004 – Chapters 17, 18.

ARCHITECTURAL CHARACTERISTICS

- Definitions and types of architectural characteristics
- Metrics for Architectural characteristics
- Fitness functions
- Component-based thinking
  - Technical approach
  - Domain driven approach

ARCHITECTURAL STYLES



ARCHITECTURAL STYLES – MORE DETAILED

- Layered
  - MVC and variants
- Pipeline
- Microkernel
- Service-based
  - Domain-driven design
- Event-driven
  - Broker
  - Mediator
- Orchestration-driven Service-Oriented architectures
  - SOAP
  - REST
- Microservices
- Volatility-driven (iDesign method)

REFERENCES

- Fundamentals of Software Architecture, Mark Richards, Neal Ford, 2020
- Righting software, Juval Lowy, 2020
- Domain Driven Design Distilled, Vaughn Vernon, Addison Wesley, 2016

DATA ACCESS

Data Access Patterns

- DAO
- Repository
- Table Module
- Active Record
- Data Mapper

Other “utility” Patterns

- Identity Map
- Lazy Load

Reference

Martin Fowler et. al, Patterns of Enterprise Application Architecture, Addison Wesley, 2003

DESIGN PATTERNS

- Creational
  - Factory Method, Prototype, Abstract Factory, Singleton
- Structural
  - Adapter, Composite, Decorator, Proxy, Bridge
- Behavioral
  - Observer, Strategy, State, Command, Chain of Responsibility

Reference

- Erich Gamma, et.al, Design Patterns: Elements of Reusable Object-Oriented Software, Addison Wesley, 1994, ISBN 0-201-63361-2

TYPES OF QUESTIONS

- In a given design recognize what principles were respected and what not. Provide a corrected design.
- Show relationships between architectural characteristics and DP (how DP support certain characteristics, or what principles are NOT applied in certain DPs).

Given a NL specification

- Identify and represent the functional requirements
- Identify and represent the architectural characteristics
- Recommend appropriate approaches ADAPTED to the problem. JUSTIFY

TYPES OF QUESTIONS

Given a specification

- Analyze different Architectural Styles for a solution and justify the most appropriate one considering **design principles** and/or **impact on architectural characteristics**
- Consider integrating different patterns in a meaningful design

Given an architecture

- Identify the used Architectural Styles and analyze their consequences.
- Describe how different ASs are related

TYPES OF QUESTIONS

Explain how different patterns relate to each other

- Ex. Strategy/State

Discuss alternative design approaches to address a specific issue (ex. data access/mapping inheritance to the data model).

Given a design

- analyze its advantages and drawbacks
- analyze its compliance to the design principles you know

QUESTION EXAMPLE

- Discuss the following code snippet considering the SOLID principles. If the case, provide a better alternative.

```
if(event instanceof PreProcess) {
    doSomething(); }
else if(event instanceof PostProcess) {
    doSomethingElse(); }
else if(event instanceof Processing) {
    doSomethingNew(); }
```

- Consider an image recognition system with the following tasks: filtering, feature extraction, labeling, clustering, scene reconstruction; assume that there is no a priori known fixed order in which these have to be called; also, the system should be easy to update with new algorithms.

i. Which architectural style is most appropriate for the given system?

ii. Give a sketch of sample architecture for the given system. Explain the interaction between the parts of the architecture.

iii. Discuss the reason for selecting a given style for the given system. Discuss possible disadvantages for the given choice.

RECOMMENDATIONS

- DO NOT present general topics!
- Answer the specific questions by giving specific solutions (i.e. give meaningful names to classes, adapt/change/integrate patterns as you need)
- Provide specific diagrams, adapted to the problem (NOT GENERAL pattern diagrams or behavior)
- Explain your design decisions showing what principles are respected and why.
- Do not just enumerate options/reproduce the theory. Apply it on the specific problem!
- Be concise, there is no minimum nr. of pages!

EXAM SETUP

- Face2face, written exam
- 3-4 subjects (as described previously)
- Exam is CLOSED books/notes. **You are not allowed to use any resources. Phones will be kept closed, inside your bags.**
- Make sure that your submitted documents contain your FAMILY_NAME and ALL FORENAMES, GROUP, YEAR OF PASSING THE LAB, LAB ASSISTANT NAME

(Ex. Popescu Ion Vasile, 30233, 2025, Alex Cosma)

Questions?

Thank you for your participation!

Good luck with your exams!

Don’t forget to send your feedback!